Exp-9--Record-IMPLEMENTATION-OF-EROSION-AND-DILATION

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Aim

To write a Python program using OpenCV to perform morphological operations such as Erosion and Dilation on an image.

The program performs the following operations:

Image Erosion Image Dilation

Software Used

Anaconda – Python 3.7 Jupyter Notebook / VS Code OpenCV (cv2) NumPy Matplotlib ##cAlgorithm Step 1: Import the required libraries: OpenCV, NumPy, and Matplotlib.

Step 2: Create a blank image using NumPy.

Step 3: Insert text onto the image using OpenCV's text drawing function.

Step 4: Display the original image.

Step 5: Create a structuring element (kernel) of suitable size.

Step 6: Image Erosion Apply the erosion operation using the created kernel. Remove pixels from the boundaries of foreground objects. Display the eroded image. Step 7: Image Dilation Apply the dilation operation using the same kernel. Add pixels to the boundaries of foreground objects. Display the dilated image. Step 8: Compare the original, eroded, and dilated images.

Program

```
Original Image
import cv2
import matplotlib.pyplot as plt
img = cv2.imread("image.png")
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title("Original Image")
plt.axis("off")
plt.show()

Erosion
kernel = cv2.getStructuringElement(cv2.MORPH_RECT, (5, 5))
erosion = cv2.erode(img, kernel, iterations=1)
plt.imshow(erosion, cmap="gray")
plt.title("Image Erosion")
plt.axis("off")
plt.show()

Dilation
```

```
kernel = cv2.getStructuringElement(cv2.MORPH_RECT, (5, 5))
dilation = cv2.dilate(img, kernel, iterations=1)
plt.imshow(dilation, cmap="gray")
plt.title("Image Dilation")
plt.axis("off")
plt.show()
```

Output:

Original Image

Image Erosion

Image Dilation

Result

Thus, the morphological operations Erosion and Dilation are successfully implemented using OpenCV.